

REVISION HISTORY			
REV	DESCRIPTION	DWN	CHK
1	ENGR RELEASE PER	AG	AG

suPI PLATFORM BOARD

CONTENTS

TITLE	SHEET
BLOCK DIAGRAM	2
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RELATED PARTS			
TYPE	NUMBER	CFG	REV
PWB	100069026	001	1
PWA	100064667	001	1
EBOM	100064667	001	1

DRAWING MEETS REQUIREMENTS OF MIL-STD-31000		CONTRACT NO. W911SR2390002	
DO NOT SCALE THIS DRAWING		DWN A. GAGE	08/23/2024
UNLESS OTHERWISE SPECIFIED CAPACITORS ARE IN MICROFARADS RESISTORS ARE IN OHMS DIMENSIONS ARE IN INCHES		CHK A. GAGE	08/23/2024
		APVD A. GAGE	08/23/2024
		APVD A. GAGE	08/23/2024
		APVD R. DEFRONZO	08/23/2024
TOLERANCE ON DECIMALS ANGLES		APVD N. BOLUS	08/23/2024
± .01 ± .1°		BY DIRECTION	DATE
± .005			

DRAPER®		555 TECHNOLOGY SQUARE CAMBRIDGE, MA 02139-3563	
PCB SCHEMATIC SUPI PLATFORM BOARD			
SIZE D	CAGE CODE 51993	DRAWING NO. 100046138	REV 1
SCALE 1:1	FILENAME: 100064667.kicad_sch		SHEET: 1/5

17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1
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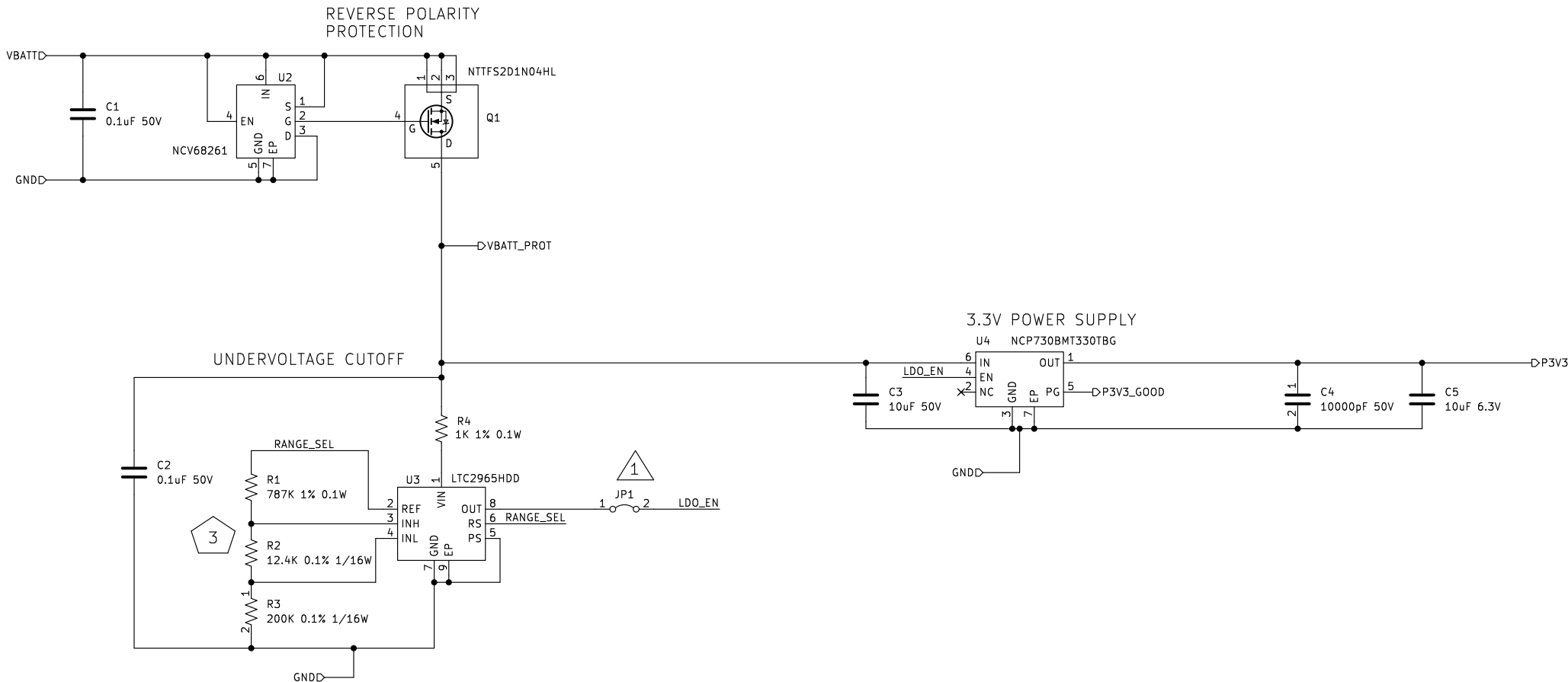
LAYOUT GUIDANCE

1 PLACE SINKSCREEN TEXT NEAR JP1:
"LO BATT CUTOFF"

DESIGN NOTES

3 RISING VOLTAGE: 20.4V
FALLING VOLTAGE: 19.2V

POWER INPUT



THIS DOCUMENT IS SUBJECT TO THE RESTRICTIONS ON SHEET 1 OF THIS DOCUMENT.			
SIZE	CAGE CODE	DRAWING NO.	REV
D	51993	100046138	1
SCALE	FILENAME: 100046138_power.kicad_sch		SHEET: 3/5
1:1	SHEET: Power Input		1

LAYOUT GUIDANCE

- PLACE JP2 AND JP3 TOGETHER.
- PLACE SILKSCREEN NEAR JP2:
"DETECT PAYLOAD"
- PLACE SILKSCREEN NEAR JP3:
"FORCE ON"
- PLACE SILKSCREEN BETWEEN JP2 AND JP3:
"SELECT"

PLACE C26 AND D1 NEXT TO J11

PLACE U8 NEXT TO J4 AND J5

PLACE R19 AND R20 NEXT TO U4

DESIGN NOTES

AUX PINS COULD BE USART1 OR I2C1 FOR COMMUNICATION WITH THE HOST PLATFORM. SERIES RESISTORS ON AUX SIGNALS ARE INTENDED TO PROTECT MICROCONTROLLER IF PINS ARE SHORTED TO GND OR 3.3V AND TO PREVENT RINGING.

THE 1-WIRE BUS CAN EITHER BE IMPLEMENTED IN FIRMWARE USING TIM1_CH4 ON PA3 OR USART2 TX/RX ON PA2/PA3.

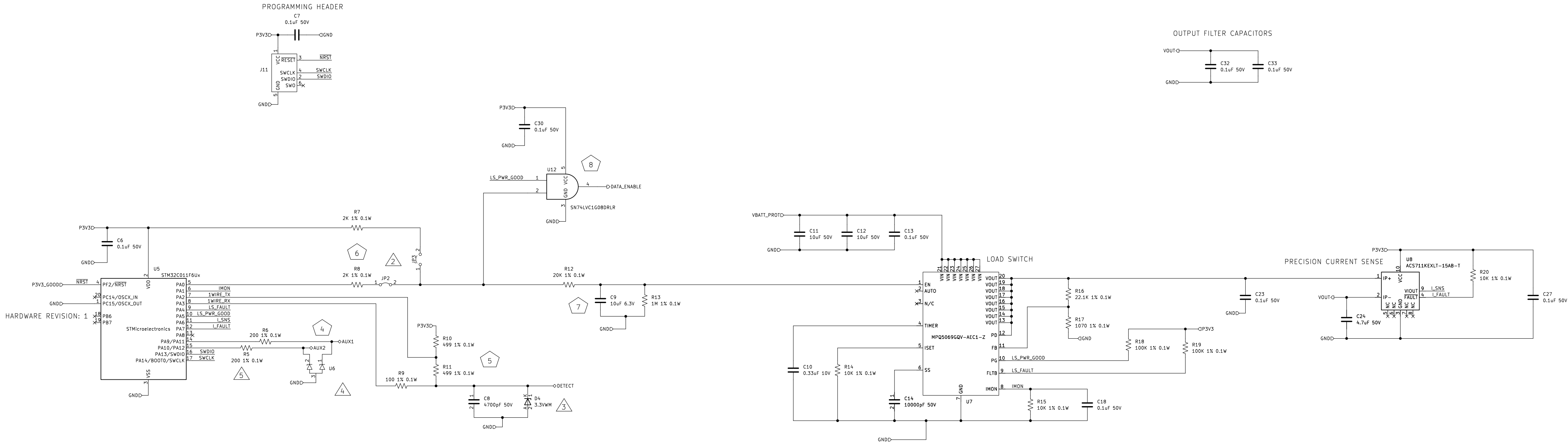
RESISTORS IN SERIES WITH THE JUMPERS ARE TO PREVENT DAMAGE TO MICROCONTROLLER IF BOTH JUMPERS ARE BRIDGED AT ONCE. ONLY ONE JUMPER IS MEANT TO BE BRIDGED AT A TIME.

RC ON ENABLE SERVES TWO PURPOSES:
1. EXTEND LOAD SWITCH OUTPUT DISCHARGE TIME. ACTIVE DISCHARGE ONLY OCCURS WHEN ENABLE LINE IS WITHIN 0.6-1.8V WINDOW.
2. PREVENT DAMAGING RINGING DUE TO LARGE INPUT CAPACITANCE

DATA SIGNAL (USB AND GPIO) ENABLE SEQUENCE:
1. ENABLE LINE ASSERTED VIA MICROCONTROLLER OR SOLDER BRIDGE
2. LOAD SWITCH ENERGIZES OUTPUT AFTER PROGRAMMED DELAY
3. POWER-ON RESET ADDS FURTHER DELAY, THEN ENABLES DATA SWITCHES
DATA SWITCHES ARE DISABLED IMMEDIATELY IF ANY OF THE FOLLOWING ARE TRUE:
1. ENABLE LINE IS DE-ASSERTED
2. LOAD SWITCH POWER GOOD IS DE-ASSERTED
3. 3.3V RAIL FALLS BELOW POWER-ON RESET THRESHOLD

PINS PC14, PC15, AND PB7 ARE RESERVED AS HARDWARE REVISION DETECTION PINS. THE FIRMWARE WILL USE THESE TO ENSURE THE CORRECT MICROCONTROLLER PIN ASSIGNMENTS.

REV	PC15	PC14	PB7
0	N/C	N/C	N/C
1	GND	N/C	N/C
2	3.3V	N/C	N/C
3	N/C	GND	N/C
4	GND	GND	N/C
5	3.3V	GND	N/C
6	N/C	3.3V	N/C
...	...	...	...
26	3.3V	3.3V	3.3V



DESIGN NOTES

- 1

DELAY AFTER DATA_ENABLE ASSERT IS 100MS MINIMUM, 200MS MAXIMUM, 160MS TYPICAL. RESET WILL ASSERT IMMEDIATELY IF DATA_ENABLE FALLS BELOW THRESHOLD.
- 2

RESET IS CONSIDERED TO HAVE LOW DRIVE STRENGTH SO IT IS EFFECTIVELY BUFFERED FOR GPIO SWITCH BY PASSING THROUGH INVERTER TWICE.

DATA SWITCHES

